ICAB March - April 2024 Exam
Case Study
Model Answer:

Draft Report

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Subject : Report on Assessment of business operation, advising on accounting & tax issues, regulatory matters, evaluation of strategic options, suggestion of investing and financing options, and analysis of ethical issues and sustainability factors of the business.

Terms of Reference:

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1. Executive Summary:

1.1. Evaluation of financial and non-financial performances of Splash Drinking Water Limited.

After incorporating the financial adjustments, financial analysis shows that Splash's revenue for the year increased substantially by 39.4% from last year due to higher sales volume and price increase. Revenue growth is much higher than the market demand growth. Revenue growth is significantly contributed by sales to sister company, Dynamic Distribution Limited (DDL) which represents over 80% of Splash's sales. Moreover, corporate sales, which is one of the key revenue streams, increased by 90% compared to last year.

Cost of revenue has also increased significantly by BDT 67,103,250 (21%). Indirect cost represents 62% of the total cost of revenue. The cost of revenue increase is mostly due to the production overhead as Splash was inefficient in limiting production overhead for marginal production.

Splash's gross profit in 2024 was BDT 576,884,460 which has increased by 55.3% from last year due to massive increase in revenue and lower increase in cost of revenue. Whereas net profit for the year was BDT 84,014,398 which has increased by 36.7% from last year. However, net profit margin for the year decreased slightly from last year.

Current ratio for the year was 0.34 which has increased in from 0.23 in prior year, but the ratio is significantly lower as compared to standard benchmark. On the other hand, quick ratio for the year is 0.28 which has slightly increased from 0.18 of last year. This indicates slight increase in Splash's ability to repay or meet the current obligation with liquid current assets. Debt to equity ratio shown increase in financial strength as it decreased to 0.72 from 1.08 in prior year. At present around 58% of Splash's capital is financed by debt and rest 42% is financed by equity. Splash was able to increase its TIE Ratio to 5.0 in current year indicating its ability to service its debts with ease.

Splash's ability to generate revenue utilizing its asset has increased as asset turnover ratio increased to 1.34 from 0.98 of last year. However, the return on sales (11.7%) is still lower than market return of 12.44%. Splash is maintaining stable profitability and achieved a return on equity of 20.1% in current year. This might be due to introduction of new filtration and purification process which is expected to reduce waste and resource consumption.

Recommendation to the board:

- Focus on reducing dependency on DDL by diversifying its customer base by exploring new retail channels or further expanding its corporate sales.
- Investigate indirect and administrative costs and implement cost-control measures to improve efficiency and profit margins.
- Prioritize on improving working capital management to enhance liquidity by exploring low-cost financing leveraging its improved financial position.

- Strengthen its internal controls and ensure full compliance with all regulatory requirements and taking actions for environmentally friendly business operation.
- Consider personalized marketing campaigns and geographical expansion for expanding the customer base, entering new regional markets or even exploring export opportunities.
- Management should properly incorporate all the necessary adjustment entry at the year end.

1.2. SWOT Analysis

Splash Drinking Water Limited has leveraged its strong association with Dynamic Group Limited to establish a solid market presence in Bangladesh, demonstrating consistent revenue growth and profitability. The company has expanded its production capacity through new facilities and partnerships with licensed bottlers, ensuring it can meet rising demand.

However, Splash faces challenges in differentiating itself due to its conventional product offerings and must innovate to cater to health-conscious and environmentally aware consumers. While the company remains profitable, careful financial management is crucial to control high operational and administrative costs. Environmental concerns, especially regarding plastic waste, have prompted Splash to plan a plastic bottle recycling initiative.

The highly regulated industry requires Splash to stay vigilant and adaptive to regulatory changes. Although DDL's extensive distribution network is a strength, the reliance on it poses a risk, making it essential to consider alternative strategies.

Diversification into emerging markets like electrolyte and alkaline water offers growth potential, and exploring international markets could open new revenue streams. Finally, enhancing digital marketing efforts is vital for reaching younger consumers and expanding brand visibility.

Recommendation to the board includes:

- Invest in the development and marketing of new product lines such as electrolyte and alkaline water and reduce dependency on traditional bottled water.
- Explore the possibility of developing an in-house distribution network or partnering with additional distributors to reduce reliance on a single entity.
- Identify potential international markets where safe drinking water is in demand and expand Splash's export operations.
- Stay ahead of regulatory developments and adapt quickly to any changes in standards or compliance requirements to mitigate potential risks.
- Continuously monitor and optimize operational processes to control costs, ensuring Splash remains competitive in a price-sensitive market.

1.3. Compatibility of strategic options with Splash's mission, vision and long-term strategies:

Splash's board is currently considering compatibility of two mutually exclusive investment options which are setting up a line for electrolyte dinks or to set up plastic water bottle recycling plant. Splash has mission of delivering pure, refreshing hydration while championing sustainability and community well-being while having a vision of becoming the leading provider of premium bottled drinking water, renowned for our commitment to purity, sustainability, and enriching lives worldwide.

Setting up the electrolyte drinking line would enable Splash to deliver pure and refreshing hydration as well as help Splash to lead the premium drinking water industry. On the other hand, setting up the recycling plant directly doesn't contribute to deliver pure and refreshing hydration, but this option will enable Splash to achieve environmental sustainability. Setting up the recycle plant will increase Splash's reputation in the industry. Both investment options show compatibility with Splash's mission and vision as both supports achievement of environmental sustainability and becoming renowned for sustainability and commitment to purity.

Recommendation to the board includes:

- As both investment options support Splash's mission and vision, board may consider either investment option to implement.
- Board should perform a thorough financial analysis and due diligence of both strategic options before making an investment decision.
- Board should also consider its risk appetite and relevant risks of the investment options.
- Board should develop a risk mitigation strategy to address operational challenges and uncertainties associated with each of the investment options.

1.4. Recommendation and justification of investment options to be executed:

Splash's board is currently considering between two mission and vision compatible investment options of 'Setting up electrolyte drinks line' and 'Setting up plastic bottle recycling plant'. To apprise the investment options, Splash's board has determined required rate of return at 15% using the Capital Asset Pricing Model (CAPM) model. Analysis shows that investment option for setting up the electrolyte drink line has a positive NPV of BDT 30,584,195 whereas the NPV of streaming service option is BDT 14,660,996. As both option is financially viable, the board should select the investment option based on higher of NPV.

Recommendation to the board:

- Splash's board should select the option for setting up the electrolyte drink like option as it has higher NPV and higher compatibility with company's mission and vision.
- Splash should perform a due diligence on prospective financial and non-financial data and conduct a thorough risk assessment, develop mitigation plan to address any potential challenges that may arise during its implementation.
- Splash management should also validate the accuracy of growth and initial outlay forecast with relevant data.

- As the other option of setting up plastic bottle recycle plant has positive NPV, Splash board may also consider this project for future investment options.

1.5. Justification of expected share premium and number of shares to be issued:

The proposed 100% share premium (10 Taka on a face value of 10 Taka) by Splash's management is supported by higher-than-average NAV (39.38 vs. 36.4) and EPS (6.53), making the premium reasonable when compared to industry peers. Given that the 100% premium is well-supported by key financial metrics, it is likely that BSEC will approve Splash's proposed premium based on NAV, EPS, and sectoral PE.

Calculations based on NAV, EPS, and sectoral PE suggest Splash's share price could range from 123 to 367 Taka, which justifies the premium. Splash's share price, based on sectoral PE, could be significantly higher than 20 Taka (including the 100% premium), indicating that Splash could potentially raise a much larger premium, up to 1000%, if desired.

Splash requires 146 million BDT over three years, and after accounting for a 3% issue cost, the total fund to be raised is 150.52 million BDT, necessitating the issuance of 7,525,773 new shares at the proposed 20 Taka per share.

Recommendation to the board:

- Splash should consider increasing its proposed premium as financial analysis suggests it could charge a much higher premium, potentially up to 1000%, given its NAV, EPS, and sectoral PE.
- Even though the 100% share premium is justifiable, Splash should prepare for potential negotiations with BSEC and be ready to provide detailed justifications based on the favourable financial comparisons to industry peers.
- Given the fund requirement over the next three years and the total number of shares needed, Splash should plan its IPO share issuance carefully to balance raising adequate capital while maintaining shareholder value.
- Splash should also consider the market sentiment and potential investor appetite when setting the final share price to ensure a successful IPO.

1.6. Determining suitability of financing through bank loan:

Splash board is considering bank loans as an alternative to financing through IPO. If Splash management plans to finance the project through bank loan, cost of capital would be 13%. Evaluation of the project feasibility with cost of bank loan would generate higher NPV compared to required rate of return.

Recommendation to the board:

- Financing the electrolyte drink project through a bank loan is feasible and advantageous.
- Lower borrowing rate compared to the CAPM rate will result in a higher NPV, further enhancing the project's profitability.
- Board may consider negotiating with bank to reduce the premium over SMART rate. Reduction in premium will reduce the borrowing rate and lead to higher profitability.

1.7. Ethical issues of Splash's business and operations:

Splash, like other similar bottled water companies, may contribute to over-extract underground fresh water, a limited natural resource for benefits while depriving society of equitable access to these resources. It needs to regularly monitor operation in order to prevent over-extraction.

Splash's appointment of A. Chowdhury & Co. (ACC) as its auditor at significantly lower fees, despite the firm's questionable reputation and previous regulatory issues, suggests a potential compromise in ethical standards. Choosing an auditor based on fee undercutting and ignoring ethical concerns could damage the company's reputation and legal standing.

The increasing popularity of bottled water, due to a lack of safe drinking water alternatives, has been met with rising prices. Bottled water manufacturers, including Splash, are exploiting consumer demand to maximize profits. Developing a socially responsible pricing model could enhance brand reputation and consumer trust.

There is a potential conflict of interest where Splash might collude with municipal authorities to maintain unsafe water supplies, increasing dependence on bottled water.

Recommendation to the board

- Splash should adopt sustainable water extraction practices, ensuring that underground water resources are conserved and available for the benefit of the wider community.
- Splash should prioritize transparency and integrity by appointing auditors with a clean and ethical track record.
- Splash should reconsider its pricing strategy for bottled water to avoid exploiting vulnerable consumers who lack access to safe drinking water.
- Splash should proactively distance itself from possible collusion with authorities and support efforts to improve municipal water safety, which would reduce unethical profiteering at the expense of public health.

1.8. Assessment of sustainability factors of Splash's business:

Positive initiatives include talent development and corporate social responsibility, enhancing human and social sustainability. Economic practices raise ethical concerns regarding resource use and pricing strategies. Environmental challenges include water depletion, plastic pollution, and

significant carbon emissions. Addressing these challenges is vital for long-term profitability and community trust.

Splash shows human sustainability efforts by creating diversity in leadership roles and the talent it acquires for bringing versatile technical knowledge and skill to the organization. Splash strictly follows the health & safety policies within the company. It has also developed operating policies to improve operational efficiency. It also creates various job opportunities which requires staffing for various positions like production, quality control, sales, marketing, and logistics.

Recommendation to the board

- Invest in water purification technologies and community initiatives that provide safe drinking water to underserved populations, reducing reliance on bottled water.
- Expand corporate social responsibility initiatives to include partnerships with local communities, ensuring that their needs and concerns regarding water extraction and environmental impacts are prioritized.
- Develop and promote programs aimed at reducing waste associated with bottled water, such as incentivizing recycling or introducing refillable options.
- Reevaluate pricing strategies to ensure fair access to water, considering the ethical implications of commodifying a basic human necessity, and invest in local community development.
- Commit to sustainable sourcing practices that protect local ecosystems and biodiversity and adopt measures to mitigate the carbon footprint associated with bottled water production and distribution.
- Foster open communication with local communities and stakeholders about water extraction practices and environmental impacts, providing avenues for feedback and collaboration to address legal and social inequalities.

2. Adjustment issues and evaluation of Splash's performances. (Requirement-a-1)

2.1. Adjustments required for current reporting and compliance issues.

Research and development cost:

During the year Splash spent BDT 50,000,000 for a research project that identified an improved filtration, purification and sterilization process than existing one and currently in development. As research outcome has led to future economic benefit, management have capitalized the research cost and reported within Other financial assets.

However, according to IAS 38, as the expenditure made for research purpose cannot be capitalized. Therefore, the BDT 50,000,000 should be charged as expense instead of capitalization. Therefore, following entry need to be made to rectify current period error:

Research expense	Dr.	50,000,000	
Other financial asset	Cr.		50,000,000

Contract with brand ambassador:

During the year Splash spent BDT 20,000,000 as part of marketing initiative by appointing 2 celebrities and influencers as brand ambassador who will serve the company for next 3 years by posting media post and participating in TV ad per year. As the TV ad and media post are yet to be done, Splash has not yet received any benefit and the expense not yet accrued. The amount paid is advance to the influences which will be allocated evenly over next three years. Therefore, marketing expense should be reversed, and an advance should be recognized.

Accordingly following adjustment is required:

Non-current advance for advertisement (20,000,000x2/3)	Dr.	1,33,33,333	
Current portion of Advance	Dr.	66,66,667	
Marketing expenses	Cr.		20,000,000

Replacement of significant component of bottling plant

Splash purchased a plant in January 2020 which had a useful life of 10 years. At the end of 2023, a major part was needed replacement. This means the part has been used for total of 4 years. The replacement part had original cost of BDT 10,000,000 whereas parts have been replaced with costs BDT 15,000,000 and added with cost of the plants. According to IAS 16, one the major part is replaced, WDV of the part need to be removed from the asset. WDV of the replaced part is 10,000,000 x (10-4)/10 = 6,000,000. Therefore, the total plant cost needs to be reduced by BDT 6,000,000. As there is no disposal value, disposed WDV of replaced part will need to be recognized as loss. Therefore, following entry need to be passed.

Loss on replacement of parts	Dr.	6,000,000	
Plant asset	Cr.		6,000,000

Interest free loan:

Splash has taken an interest free loan of BDT 3,000,000 for 3 years from its parent company DGL. Annual repayment instalment is BDT 1,000,000. The loan which is a financial liability has not been subsequently measured at amortized cost in line with IFRS 9. The annual instalment needs to be discounted with the incremental borrowing rate. Incremental borrowing rate for Splash is SMART +3.5% which is 9.5%+3.5%= 13%.

Considering the instalment is payable at the year end, the present value of the loan therefore is:

$\frac{1,000,000}{(1+0.13)^1} + \frac{1,000,000}{(1+0.13)^2} + \frac{1,000,000}{(1+0.13)^3}$ = BDT 2,361,153

Difference between loan amount received and present value of the loan is BDT 638,847 which should be recognized as additional equity contribution by parent company and reduced from loan amount. Furthermore, an interest expense for the year should be charged on the present value of the loan at the rate of 13% (2,361,153 x 13%). Accordingly following entry should be made:

Intercompany loan	Dr.	638,847	
Additional equity component	Cr.		638,847
Interest expense	Dr.	306,950	
Intercompany loan	Cr.		306,950

VAT Audit by authority:

During a VAT inspection, VAT authority identified that Splash have not deducted VAT of Tk 10,000,000 from its suppliers. Subsequently Splash made the payment and recorded the payment as Advance to suppliers. The amount is not recoverable from the suppliers. Therefore, these advance need to be written off. Therefore, following entry is required:

Loss on non-deduction of VDS	Dr.	10,000,000	
Advance to suppliers	Cr.		10,000,000

2.2. Evaluation of financial after adjustment

Splash's financial statement has been adjusted for the reporting and compliance issues mention by the Finance Director. After making necessary adjustments to the financial statements, financial analysis has been performed on the Splash's adjusted financial statements.

Revenue:

There has not been any impact of the adjustment on Revenue. Splash's revenue for the year is BDT 963,116,460 which has increased substantially by 39.4% (BDT 272,413,260) from last year revenue of BDT 690,703,200. This growth is largely attributed to higher sales volume through different sales channels as well as price increase following the market trend. Furthermore, the increase in revenue is much higher than market demand for bottled water which is growing at a rate of 15-20% annually in Bangladesh. However, in absence of exact price change of Splash bottled waters, its effect on revenue increase could not be properly ascertained.

More than 80% of Splash's sales are earned through its sister company Dynamic Distribution Limited. During the year Splash's sales to DDL increased from 597,458,268 to 780,124,333. Sales to DDL has increased by 31% and contributed BDT 182,666,065 in the sales growth which represents 67% of the growth factor.

Another important source of revenue for Splash is sales to corporate businesses. During the year sales to corporate increased to BDT 144,467,469 which is up by 90% from last year corporate sales of BDT 75,977,352. This matches with the post-Covid trend as employees of most corporates are returning to office and number of physical meetings has increased. Sales to corporate house has contributed 25% of revenue growth.

Although private label sales represent less than 5% of Splash's overall sales, it has contributed 8% in sales growth. Despite being the smallest source of revenue stream, private level sales have experienced the highest sales growth as it has increased by 123% from 17,267,580 to 38,524,658. This was caused by gradual increase in demand for private label water bottled by hotels and restaurants.

Cost of Revenue:

Cost of revenue has increased significantly by 21% (BDT 67,103,250) leading to BDT 386,232,000 in 2023 from BDT 319,128,750 in 2022. The high increase of cost of revenue is due to massive increase in Revenue. However, the cost of revenue increase rate is not entirely matching with revenue increase rate. This could be due to increase in product price which doesn't contribute to cost of revenue increase.

Cost of revenue consists of prime cost and indirect costs. Prime cost represents 38% of total cost of revenue which has slightly decreased from 41% of last year. On the other hand, indirect cost represents 62% of the cost of revenue in the current year which was 59% of last year.

During the year cost of revenue has increased by BDT 67,103,250. Of this amount 76% (BDT 51,177,878) has been contributed by increase in indirect cost whereas remaining 24% (BDT 15,925,372) has been contributed by prime cost. This clearly shows that increase in cost of revenue has been increased mostly due to other indirect costs. This shows that the production cost for increased water bottle quantity is not that significant for increase in cost of revenue. However, the cost of revenue increase is mostly due to the production overhead costs. This indicates that Splash might not be efficient enough to manage the production overhead costs for increased production quantity.

GP and GP Margin:

During the year Splash has achieved gross profit of BDT 576,884,460 which has increased by 55.3% (BDT 205,310,010) from last year gross profit of BDT 371,574,450. This increase in gross profit resulted from massive increase in revenue and comparatively lower increase in cost of revenue.

Splash's profit margin for the year has been determined at 59.9% which has been 53.8% in the previous year. This shows that there has been impressive growth in the GP margin which is mostly due to growth in revenue.

Net profit and Net profit margin

After adjustment of the accounting issues, Splash has achieved net profit of BDT 84,014,398 which has increased by BDT 22,561,793 from previous year net profit of BDT 61,452,605. This has been increased significantly by 36.7% which has been resulted due to massive growth in revenue, moderate increase in cost of revenue. Non-operating income also contributed to increase in Splash's profitability.

Net profit margin for the year is 8.7% which has slightly decreased from last year net profit margin of 8.9%. This indicates that overall profitability for Splash is pretty consistent. However, this is surprising as Net profit margin has decreased despite there were massive growth in revenue and the market is showing positive trends. One of the reasons for decrease in profit margin is administrative expense has almost doubled during the year. It is apparent that Splash could not control its cost.

Current and Quick Ratio

Analysis of adjusted financial statement shows that Current ratio of Splash is 0.34 which has increased from 0.23 of the previous year. Even though Splash has increased its current ratio from previous year, still the ratio is significantly lower. Splash's lower current ratio indicates that current asset is much lower comparing to the current liability. Furthermore, it has experienced an increase in income tax liability which has contributed to lowering the current ratio. Splash's current ratio is lower due to low amount of inventory and receivables which should have been much higher for a manufacturing company. Current year ratio has increased due to increase in inventory, receivables and advance.

Splash's quick ratio has followed the Current ratio's trend. Its quick ratio for the year is 0.28 which has slightly increased from 0.18 of prior year. Quick ratio indicates ability of an entity to repay or meet the current obligation with current assets which can be liquidated on quickest time. From analysis of Splash's financial statements, it appears that Splash can only convert and pay off 28% of current obligation with its current assets excluding inventories. However, as the quick ratio in the current year has increased slightly from prior year, this increase signifies that quickly liquifiable asset including cash and cash equivalent has increased.

Debt to Equity and TIE Ratio

Debt to equity ratio shows how much debt an entity has compared to its assets. Analysis of Splash's adjusted financial statement shows that it has debt to equity ratio of 0.72 in 2023 which has decreased from 1.08 of last year. Debt to equity has decreased due to increase in retained earnings and decreased non-current liabilities like long-term borrowing and lease liabilities. At present nearly 58% of its capital is financed by debt and remaining 42% is financed by equity. This has improved from last year where the equity represented 48% and liability represented 52% in the capital structure. This clearly indicate that Splash has strong equity position arising from profit reserve and declining trend in non-current liabilities. Splash's gearing is still at lower side. It still has room to increase gearing and obtain tax benefit from tax shield.

Times interest earned ratio (TIE) is a solvency ratio indicating the ability to pay all interest on business debt obligations. Splash's TIE Ratio has increased to 5.0 in current year from last year TIE

ratio of 3.9. This indicates the Splash's ability to generate profit compared to finance cost has increased significantly during the year. Increase in TIE ratio in 2023 is because of decreasing finance cost and increased profitability during the year. Splash has paid off some portion of its interest-bearing loan during the year which led to decrease in finance cost. High TIE ratio indicates that Splash has the ability to serve its debts 5 times with its current level of profitability. This clearly shows Splash's strong financial position and performance.

Asset Turnover and Return on Assets

Asset turnover ratio shows an entity's efficiency of generating revenue through utilizing its assets. For Splash, Asset turnover ratio in 2023 is 1.34 which slightly increased from 0.98 of previous year. This shows that Splash is able to generate revenue utilizing its asset has increased from prior year. This is in line with post-covid business trend which is showing positive signs. This shows that Splash has increased its efficiency in asset utilization and revenue generation. As demand for bottled water is increasing every year, there is chance for Splash to increase its revenue without spending much on asset.

Return on asset indicates the profit percentage generated by management by using entity's resources. Return on asset in 2023 was calculated at 11.7% which shows a smart increase from 8.8% of last year return on assets. Although there has been increase during the year, but the return rate is lower than market return of 12.44%. Even the risk-free SMART rate is 9.5% which is increasing every six months. Considering this Splash has sufficient room for improvement of return against resources deployed.

Return on Equity

During the year Splash has generated return of equity of 20.1% which was 18.2% in last year. This shows that Splash has been consistently generating a stable profit. Furthermore, Splash has witnessed a smart growth in the current year despite growth in equity. The increase in return on equity is due to increase in revenue and profitability during the year. This shows that Splash has performed well by increasing and generating a handsome return for the shareholders. Furthermore, it shows Splash's efficiency in financial management and maximizing shareholders wealth.

2.3. Evaluation of non-financial performance

Similar to financial performance, Splash has shown outstanding non-financial performance during the year.

Environmental Responsibility

Splash has made strides in addressing environmental concerns. Although the company operates in an industry scrutinized for its environmental impact, it has taken steps to mitigate its footprint. The new filtration and purification process is expected to reduce waste and resource consumption. However, there is still an opportunity for Splash to further enhance its environmental responsibility by exploring sustainable packaging options and reducing plastic waste.

Regulatory Compliance

Regulatory compliance remains a critical aspect of Splash's non-financial performance. The recent VAT audit revealed gaps in compliance, particularly in VDS deductions. Splash has taken steps to address these issues and enhance internal controls, reflecting its commitment to good governance.

Technological Advancements

Splash has shown a strong commitment to technological advancements. The company has invested in cutting-edge technology to improve its operations and product offerings. These technological investments not only enhance product quality but also contribute to operational efficiency and customer satisfaction.

Customer Experience

Enhancing the customer experience has been a focal point for Splash. The company's efforts to improve product quality through advanced purification processes directly impact customer satisfaction. Additionally, Splash's promotional strategies and brand ambassador initiatives help creating a positive brand image and fostering customer loyalty.

Operational Efficiency

During the year, Splash Drinking Water Limited significantly improved its operational efficiency. The company implemented a new research project that resulted in an enhanced filtration, purification, and sterilization process. This advancement not only improved the quality of the water but also optimized resource management, demonstrating Splash's commitment to operational excellence and efficiency. This improved process likely contributed to the company's overall performance and customer satisfaction.

2.4. Conclusion

- Splash's revenue for the year increased substantially by 39.4% from last year due to higher sales volume and price increase.
- Increase in revenue is much higher than market demand which is growing at a rate of 15-20% annually in Bangladesh.
- Over 80% of Splash's sales are through its sister company, Dynamic Distribution Limited (DDL). While this has contributed significantly to revenue growth, such heavy reliance poses a risk.
- During the year, corporate sales increased 90% higher than last year.
- Although private label sales represent less than 5% of Splash's overall sales, its demand is gradually increasing year on year.
- Cost of revenue has increased significantly by 21% (BDT 67,103,250) matching the substantial increase in Revenue.
- Indirect cost represents 62% of the cost of revenue which has increased from 59% of last year.
- The cost of revenue increase is mostly due to the production overhead due to Splash's inefficiency in limiting production overhead for marginal production.

- Splash/ gross profit of BDT 576,884,460 in 2024 increased by 55.3% from prior year due to massive increase in revenue and lower increase in cost of revenue.

- Splash has achieved net profit of BDT 84,014,398 which has increased significantly by 36.7%.

- Net profit margin for the year decreased slightly from last year.

- Even though current ratio of Splash is 0.34 which has increased in from 0.23 in prior year, but the ratio is significantly lower than standard.

- Quick ratio for the year is 0.28 which has slightly increased from 0.18 of prior year which indicates slight increase in Splash's ability to repay or meet the current obligation with liquid current assets.

- Debt to equity ratio of 0.72 decreased from 1.08 in prior year due to increase in retained earnings and decreased in non-current liabilities like long-term borrowing and lease liabilities.

- Nearly 58% of its capital is financed by debt and rest 42% is financed by equity which clearly indicate that Splash has strong equity position.

- Splash's TIE Ratio has increased to 5.0 in current year from prior year indicating increased ability to generate profit compared to finance cost.

- Splash's ability to generate revenue utilizing its asset has increased as asset turnover ratio increased to 1.34 from 0.98 of previous year.

- Although the return on asset has been calculated at 11.7%, but the return rate is lower than market return of 12.44%.
- Splash has a stable profitability which is indicated by return of equity for the year is 20.1% and 18.2% in prior year.

- Splash introduced the new filtration and purification process which is expected to reduce waste and resource consumption.

- Splash has invested in cutting-edge technology to improve its operations and product offerings.

- Splash implemented a new research project that resulted in an enhanced filtration, purification, and sterilization process. This not only improved the quality of the water but also optimized resource management.

2.5. Recommendation

- With administrative expenses almost doubling, Splash should evaluate these costs to identify and eliminate unnecessary expenditures, thus improving net profit margins.

- Splash should focus on diversifying its customer base to reduce dependency on DDL, potentially exploring new retail channels or expanding its corporate sales further.

- Splash should investigate the sources of indirect costs and implement cost-control measures, particularly in production overheads, to improve efficiency.

- Splash's liquidity position is weak. It should work on improving its working capital management, possibly by optimizing inventory levels, improving receivables collection, or renegotiating payment terms with suppliers to enhance liquidity.
- Splash should focus on better asset utilization or strategic investments that can yield higher returns on assets, aligning more closely with market expectations.
- Splash can consider leveraging its improved financial position by exploring low-cost financing options. This could help in funding expansion or modernization initiatives without significantly increasing financial risk.
- Splash should take further steps in environmental responsibility, particularly by exploring sustainable packaging solutions to reduce plastic waste.
- Splash should continue to strengthen its internal controls and ensure full compliance with all regulatory requirements.
- Splash should consider implementing customer feedback loops and loyalty programs to deepen customer engagement and retention. Personalized marketing campaigns could also help in expanding the customer base.
- Splash should consider geographical expansion, possibly entering new regional markets or even exploring export opportunities.
- Management should properly incorporate all the necessary adjustment entry at the year end.

3. Evaluation of the Strengths, Weaknesses, Opportunities, and Threats (SWOT) of Splash's existing business (Requirement a-ii):

3.1. SWOT Analysis:

Strengths

- ***Strong Brand Association with Dynamic Group Limited (DGL):***
  Splash benefits significantly from its association with DGL, a large conglomerate in Bangladesh. This association has transferred consumer trust and preference from DGL to Splash, aiding in rapid brand recognition and growth.

- ***Expanding Production Capacity:***
  Splash has steadily expanded its production capacity by setting up additional factories and collaborating with licensed bottling plants. This expansion allows Splash to meet the increasing national demand, positioning it well to penetrate and make mark in the entire market.

- ***Comprehensive Distribution Network:***
  Splash utilizes Dynamic Distribution Limited's (DDL) extensive distribution network, ensuring wide availability across Bangladesh. This allows efficient and effective distribution of products nationwide.

- ***Focus on Quality and Safety:***
  Splash adheres to stringent quality control measures, ensuring that its bottled water meets the Bangladesh Standards and Testing Institution (BSTI) standards for drinking water. This focus on quality builds consumer trust in the product.

- ***Diverse Product Range:***
  Splash offers a wide variety of bottle sizes, catering to different consumer needs from individual consumption to large social gatherings, making its product accessible to a broad market segment.

- ***Financial Stability:***
  Splash has shown strong financial performance, with consistent revenue growth and profitability, reflecting effective management and market penetration strategies.

- ***Efficient Cost Management:***
  Through strategic outsourcing and in-house production, Splash has managed to control its costs effectively, ensuring competitive pricing in a crowded market.

- ***Commitment to Sustainability:***
  Splash's mission emphasizes sustainability, and the company is exploring environmentally friendly initiatives like setting up a plastic bottle recycling plant, which can enhance its brand image.

Weakness

- ***High Operating Costs:***
  The company faces significant administrative and operational expenses, which have the potential to erode profit margins if not managed effectively.

- ***Limited Brand Differentiation:***
  Despite being a well-known brand, Splash faces challenges in differentiating itself from other local and international competitors in the bottled water market.

- ***Inconsistent Water Source Quality:***
  The reliance on different water sources, including WASA-supplied water, can lead to variability in water quality, which could impact consumer perception and trust.

- ***Capacity Constraints in High Demand Periods:***
  Although Splash has expanded its production capacity, demand spikes during peak seasons could still lead to supply shortages, affecting market share.

- ***Limited Innovation in Product Offerings:***
  While the company is exploring new product lines like electrolyte water, its current product range lacks significant innovation, which may limit its appeal to younger or more health-conscious consumers.

- ***Dependency on Licensed Bottling Plants:***
  The use of third-party bottling plants, while cost-effective, could lead to quality control issues if these partners do not adhere to Splash's stringent standards.

- ***Exposure to Regulatory Risks:***
  The bottled water industry is highly regulated, and any changes in standards or compliance requirements could increase operational costs or limit market access.

Opportunities

- ***Expansion into Electrolyte and Alkaline Water Market:***
  The growing demand for functional beverages like electrolyte and alkaline water presents an opportunity for Splash to diversify its product line and capture a new market segment.

- ***Increased Demand for Sustainable Products:***
  With rising consumer awareness about environmental issues, there is an opportunity for Splash to lead in sustainability by introducing eco-friendly packaging and promoting its planned recycling initiatives.

- ***Growing Urbanization and Disposable Incomes:***
  As urbanization increases in Bangladesh, so does the demand for convenient, safe drinking water, offering Splash an expanding market base.

- ***Export Market Expansion:***

With its established production capacity, Splash has the potential to expand its footprint in international markets, particularly in regions where safe drinking water is a concern.

- ***Leveraging Digital Marketing:***
  The growing trend of eCommerce and digital marketing provides Splash an opportunity to reach a wider audience and increase brand engagement through online platforms.

- ***Partnerships with High-End Restaurants:***
  By expanding its private-label bottled water production for restaurants, Splash can establish a premium brand image and increase its market penetration.

- ***Health-Conscious Consumer Trends:***
  With the rising trend of health and wellness, Splash can market its water as a healthier alternative to sugary drinks, appealing to a broader demographic.

- ***Regulatory Support for Environmental Initiatives:***
  Government incentives for businesses that promote environmental sustainability can be leveraged to support Splash's recycling plant and other green initiatives.

Threats

- ***Overreliance on DGL's Distribution Network:***
  While the use of DDL's distribution network is a strength, it also represents a dependency that could become a vulnerability if any issues arise within DDL.

- ***Intense Competition:***
  The bottled water market in Bangladesh is highly competitive, with over 30 brands vying for market share, including major multinational companies with significant resources.

- ***Environmental Concerns:***
  The growing backlash against single-use plastics and the environmental impact of bottled water production poses a significant threat to the industry. This could lead to stricter regulations or a shift in consumer preference away from bottled water.

- ***Water Source Depletion:***
  The over-extraction of groundwater, particularly in areas where Splash operates, could lead to sustainability issues, affecting long-term production capabilities.

- ***Regulatory Compliance Risks:***
  Changes in regulations regarding water quality, packaging, and environmental impact could impose additional costs or limit the company's operational flexibility.

- ***Price Sensitivity of Consumers:***
  The economic situation in Bangladesh makes consumers highly price-sensitive, and any increase in production or distribution costs that lead to higher prices could reduce demand.

- ***Potential Supply Chain Disruptions:***
  Disruptions in the supply chain, whether from raw material shortages, logistical issues, or partner failures, could impact Splash's ability to meet market demand. This could lead to reputational damage and loss of business.

- ***Reputational Risk:***
  Any quality control issues, such as contamination of water or use of substandard bottles, could lead to a loss of consumer trust and long-term reputational damage.

- ***Economic Instability:***
  Fluctuations in the macroeconomic environment, including inflation, changes in disposable income, and currency instability, could affect consumer spending on bottled water. This in turn could reduce Splash's sells.

3.2. Conclusion:

- Splash Drinking Water Limited has established itself as a key player in the Bangladeshi bottled water market, by consistently generating revenue growth and profitability utilizing the strong brand association with Dynamic Group Limited.

- Splash has made significant expansion of its production capacity through new factories and partnerships with licensed bottling plants. This expansion allows the company to meet the increasing demand and ensures that it can maintain a strong market presence.

- Splash is facing challenges in differentiating itself from competitors due to similar and conventional product offerings. There is a need for more innovation to cater to evolving consumer preferences, particularly in health-conscious and environmentally aware segments.

- Although the company remains profitable, careful financial management is necessary to prevent high operational and administrative costs from eroding profit margins, especially in a competitive and price-sensitive market.

- The bottled water industry is under increasing scrutiny for its environmental impact, particularly concerning plastic waste and water source depletion. Splash's plan to set up a plastic bottle recycling plant is a step toward right direction.

- The bottled water industry in Bangladesh is highly regulated, and any changes in regulatory standards could have significant implications for Splash. The company must remain vigilant and adaptive to ensure compliance and mitigate potential risks associated with regulatory changes.

- Splash's reliance on Dynamic Distribution Limited (DDL) for distribution is both a strength and a vulnerability. While DDL's extensive network ensures wide product availability, this dependency could become a bottleneck if issues arise within DDL, making it critical for Splash to consider alternative distribution strategies.

- There is substantial growth potential in diversifying into new product lines like electrolyte and alkaline water which are gaining popularity among health-conscious consumers. Entering

these segments early could allow Splash to capture significant market share in these emerging categories.

- The company is well-positioned to explore international markets, particularly in regions where access to safe drinking water is limited. Expanding its export operations could provide a new revenue stream and reduce dependency on the domestic market.

- In a rapidly digitalizing market, leveraging digital marketing and eCommerce platforms is essential. Splash should enhance its online presence to reach younger, tech-savvy consumers, which would complement its traditional distribution channels and increase brand visibility.

3.3. Recommendation:

- Invest in the development and marketing of new product lines such as electrolyte and alkaline water to attract health-conscious consumers and reduce dependency on traditional bottled water.

- Expedite the development of the plastic bottle recycling plant and promote eco-friendly packaging to align with growing consumer demand for sustainable products.

- Implement more rigorous quality control measures across all bottling plants, including third-party facilities, to ensure consistent product quality.

- Explore the possibility of developing an in-house distribution network or partnering with additional distributors to reduce reliance on a single entity.
- Leverage digital marketing and eCommerce platforms to expand consumer reach, particularly among younger, tech-savvy demographics.

- Identify potential international markets where safe drinking water is in demand and expand Splash's export operations.

- Stay ahead of regulatory developments and adapt quickly to any changes in standards or compliance requirements to mitigate potential risks.

- Continuously monitor and optimize operational processes to control costs, ensuring Splash remains competitive in a price-sensitive market.

4. Compatibility of investment options with Splash's Mission and Vision (Requirement B-i):

Splash's board is currently evaluating between two mutually exclusive investment options "setting up line for electrolyte drink" or "setting up plastic bottle recycling plant". Board is also considering the compatibility of these investment options with Splash's mission and vision.

4.1. Compatibility with Splash's Mission:

Splash's mission is "to deliver pure, refreshing hydration while championing sustainability and community well-being". If splash selects to set up line for electrolyte drink, it will be able to deliver pure and refreshing hydration to the community. Consumers who are leading active lifestyle will be

benefited from consuming these healthy electrolyte drinks. These electrolyte drinks will help the consumers to regain the electrolytes lost during physical activities. In that sense, the electrolyte drinks will also take care of community well-being.

Other investment option is to set up plastic bottle recycling plant. If the recycling plant is set up, Splash could recycle its own plastic bottles as well as other company's plastic bottles. This would help the environment and increase industry sustainability. Although recycling water bottle will not lead to refreshing hydration, but it will support its mission to champion sustainability and community well-being.

4.2. Compatibility with Splash's Vision:

Splash's vision is 'to become the leading provider of premium bottled drinking water, renowned for our commitment to purity, sustainability, and enriching lives worldwide'. This means Splash wants to lead the premium drinking water industry which is known for purity and sustainability. By setting up the electrolyte drinking line, it will be able to take a step forward in becoming the leader in premium drinking water industry as well as increase its reputation for purity. On the other hand, by setting up the bottle recycling plant, Splash could increase its reputation by becoming a business promoting environmental sustainability.

4.3. Conclusion:

- Splash has mission to deliver pure, refreshing hydration while championing sustainability and community well-being.
- Splash's vision is to become the leading provider of premium bottled drinking water, renowned for strong commitment to purity, sustainability, and enriching lives worldwide.
- Splash's board is currently considering compatibility of two mutually exclusive investment options which are setting up new line for electrolyte drinks or to set up bottle recycling plant.
- Setting up the electrolyte drinking line would enable Splash to deliver pure and refreshing hydration as well as help Splash to lead the premium drinking water industry. Therefore, this investment option shows compatibility with Splash's mission and vision.
- Setting up the recycling plant directly doesn't contribute to deliver pure and refreshing hydration, but this option will enable Splash to achieve environmental sustainability. Setting up the recycle plan will increase Splash reputation in the industry. Therefore, this investment option will indirectly support achieving Splash's mission to achieve sustainability and vision of becoming renowned for sustainability and commitment to purity.

4.4. Recommendation:

- As both investment option supports Splash's mission and vision, board may consider either investment option to implement.
- Board should perform a thorough financial analysis and due diligence of both strategic options before making an investment decision.
- Board should also consider its risk appetite and relevant risks of the investment options.

- Board should develop a risk mitigation strategy to address operational challenges and uncertainties associated with each of the investment options.

5. Recommendation with justification of investment options to be executed (Requirement b-ii):

Splash's board is currently considering between two mutually exclusive investment options. First option is to set up a new line for electrolyte drinks and second option is to set up a water bottle recycling plant. Each strategic option is compatible with Splash's mission, vision and long-term strategies to achieve mission and vision. As both investment options are mutually exclusive, board need to select and implement the option which has more financial viability and favourable due diligence feedback.

Both investment options are new to Splash and has little knowledge and experience. Both options will require significant cash outflow at the beginning and expected to generate good revenue over the year. Therefore, investment decision should be taken based on return on investment and Net Present Value (NPV) analysis.

Splash's board apprises the investment options using Capital Asset pricing Model (CAPM). From the information provided, Risk Free return (SMART rate) is 9.5%, Market Return is 12.44% and market beta is 1.53. Considering is CAPM rate is calculated at 9.5%+ 1.53 x (12.44%-9.5%) = 9.5% + 4.5% = 14%. NPV of both options are analysed below using the CAPM rate.

5.1. NPV of option to set up electrolyte drink line:

Splash expects to generate Revenue of BDT 40,000,000 in the first year which will have expected growth of 70% in the second year, 50% in third year, 30% in the fourth year and 20% in the fifth year.

The investment option will require an initial outlay of BDT 110,000,000 for the plant and further BDT 10,000,0000 as initial advance for renting out the production facility. New electrolyte dinks are expected to have 40% cost of revenue with annual outflow for administrative expenses of BDT 40,000,000 and distribution cost equivalent to 10% revenue. The tax rate is expected to remain stable at 25% on taxable profit. However, due to minimum taxation, no benefit will be available on carried forward loss. Cash flows are expected to grow 3% perpetually after fifth year.

From the forecast of finance team, following cashflows can be generated which will be discounted at 14%.

Cash flows	**Year 0**	**Year 1**	**Year 2**	**Year 3**	**Year 4**	**Year 5**
Initial investments	(120,000,000)					
Revenue		40,000,000	68,000,000	102,000,000	132,600,000	159,120,000
Cost of revenue		(16,000,000)	(27,200,000)	(40,800,000)	(53,040,000)	(63,648,000)
Distribution exp		(4,000,000)	(6,800,000)	(10,200,000)	(13,260,000)	(15,912,000)
Admin Expense		(40,000,000)	(40,000,000)	(40,000,000)	(40,000,000)	(40,000,000)
Cash flow before tax	**(120,000,000)**	**(20,000,000)**	**(6,000,000)**	**11,000,000**	**26,300,000**	**39,560,000**

Tax at 25%	-	-	-	(2,750,000)	(6,575,000)	(9,890,000)
Free Cash flow	**(120,000,000)**	**(20,000,000)**	**(6,000,000)**	**8,250,000**	**19,725,000**	**29,670,000**
Terminal value *						269,727,273
Total Free Cash flow	**(120,000,000)**	**(20,000,000)**	**(6,000,000)**	**8,250,000**	**19,725,000**	**299,397,273**
PVIF @ 14%	1.00	0.88	0.77	0.67	0.59	0.52
PV	**(120,000,000)**	**(17,543,860)**	**(4,616,805)**	**5,568,515**	**11,678,783**	**155,497,562**
NPV						**30,584,195**

* Terminal value = Cash flow at Y5/(d-g) where d is discount rate and g is growth rate.
= 29,670,000/ (0.14-0.03) = 269,727,273

Based on the calculation, it is expected that setting up electrolyte drink line will generate NPV of BDT 30,584,195. As the NPV of the investment option is positive, it may be accepted.

5.2. NPV of option to setting up plastic bottle recycling plant:

Splash management has forecasted that recycling plant will cost BDT 75,000,000 for the plant and further BDT 5,000,0000 as initial advance for renting out the facility. Furthermore, annual outflow of BDT 25,000,000 will be required for operating expenses including rent, salary, sterilizing process, and maintenance of the facility. It is expected that the plastic bottles that will be recycled can be obtained for free, but BDT 15,000,000 will be spent annually for collecting the plastic bottles and transporting them to the recycling facility. From the recycled bottles, we expect to generate 50,000,000 in first two years either by saving expense of bottle purchase or selling the recycled bottles to others. The cost saving and income would increase to BDT 55,000,000 per annum for next three years. Management expects to will obtain 50% tax exemption for our environmental conservation effort. and 2% perpetually growth after the fifth year. Based on the information, following cash flows can be derived:

Cash flows	Year 0	Year 1	Year 2	Year 3	Year 4	Year 5
Initial investments	(80,000,000)					
Income		50,000,000	50,000,000	55,000,000	55,000,000	55,000,000
Annual expenses		(25,000,000)	(25,000,000)	(25,000,000)	(25,000,000)	(25,000,000)
Collection & transportation		(15,000,000)	(15,000,000)	(15,000,000)	(15,000,000)	(15,000,000)
Cash flow before tax	**(80,000,000)**	**10,000,000**	**10,000,000**	**15,000,000**	**15,000,000**	**15,000,000**
Tax @12.5% *		(1,250,000)	(1,250,000)	(1,875,000)	(1,875,000)	(1,875,000)
Free Cash flow	**(80,000,000)**	**8,750,000**	**8,750,000**	**13,125,000**	**13,125,000**	**13,125,000**
Terminal value ^						109,375,000
Total Free Cash flow	(80,000,000)	8,750,000	8,750,000	13,125,000	13,125,000	122,500,000
PVIF @ 14%	1.00	0.88	0.77	0.67	0.59	0.52
	(80,000,000)	7,675,439	6,732,841	8,859,001	7,771,054	63,622,661
						14,660,996

* Tax rate is 25%. After 50% exemptions, tax rate is calculated at 12.5%.

^ Terminal value = Cash flow at Y5 where d is discount rate and g is growth rate.
= 13,125,000/ (0.14-0.02) = 109,375,000

Based on the calculation, setting up plastic water bottle recycling plant option has positive NPV of BDT 14,660,996. Therefore, this option is also commercially viable.

5.3. Suggestion on investment option with justification:

Between the two investment options, Splash board should select the one which is commercially viable and compatible with its mission and vision. From this perspective, both investment options "Setting up electrolyte drink line" and "Setting up plastic bottle recycling plant" has positive NPV and matches with Splash's mission and vision. However, setting up the electrolyte line will generate higher NPV (BDT 30,584,195) for Splash. Furthermore, this investment option is closely related with Splash's mission of providing pure & refreshing hydration as well as its vision to be the leader in drinking water industry. Therefore, selecting to set up electrolyte dink line option is suggested and more justified.

5.4. Conclusion:

- Splash's board is currently considering between two investment options each of which are compatible with Splash's mission and vision.
- Splash's board apprises investment options with the rate of return determined using the Capital Asset Pricing Model (CAPM) which is derived at 14%.
- Using the Required rate of return of 14% and considering 3% perpetual growth, cash flows for setting up electrolyte drink line generates a positive NPV of BDT 30,584,195. This indicates that this investment option is financially viable.
- Considering 14% required rate of return and 2% perpetual growth, cash flows for plastic bottle recycle plant option generates a positive NPV of BDT 14,660,996. This also suggest that setting up recycle plant is also financially viable.
- As the investment options are mutually exclusive, Splash board should consider electrolyte drink line option as it has higher NPV.

5.5. Recommendation:

- Based on the NPV analysis, Splash's board should select the option for setting up the electrolyte drink line option as it has higher NPV. Furthermore, this option is directly compatible with Splash's mission of providing refreshing hydration and vision of leading the industry with reputation.
- Splash should perform a due diligence on prospective financial and non-financial data and conduct a thorough risk assessment, develop mitigation plan to address any potential challenges that may arise during its implementation.
- Splash management should also review the accuracy of revenue and its growth forecast and validate the source with relevant data.
- Splash board should also review the directly attributable cost structure as well as initial outlay during due diligence.
- As the other option of setting up plastic bottle recycle plant has positive NPV, Splash board may also consider this project for future investment options.

6. Justification of share premium and number of shares to be issued (Requirement b-iii):

6.1. Justification of 100% Share premium rate for BSEC approval:

Splash's management is expecting to raise fund from IPO with 100% share premium i.e. 10 Taka premium with per share with Face Value of 10 Taka. However, board is not certain whether the premium will be approved by the Bangladesh Securities & Exchange Commission (BSEC). BSEC approves the premium after having equal consideration on NAV, EPS and sectoral PE ration.

Based on the provided data from comparable companies, average NAV, Price and PE is below:

SL.	Company Name	NAV on February 2024 (BDT)	3-month average market price (BDT)	EPS on February 2024 (BDT)	PE (Price/EPS)
1	Dhaka Consumer Products PLC	16	35	1.56	22.43
2	Crystal Beverage PLC	50	150	11.50	13.04
3	Fresh Foods PLC	13	56	1.21	48.28
4	Standard Drinking Water PLC	14	65	1.30	50
5	Continental Food Processing PLC	89	264	7.17	36.82
Average		**36.40**	**114.00**		**34.11**

The draft financial statement of Splash shows that Splash has NAV of BDT (765,625,913/ 19,441,500) = 39.38 whereas the EPS is (126,898,025/ 19,441,500) = 6.53.

This shows that Splash has higher NAV than average NAV of 36.4 from comparable companies. Average share price of these companies is 114 taka which shows that average PE ratio is 34.11.
Based on the NAV of the comparable companies, Splash's share price could be (114*39.38/36.4) = 123.
Based on Sectoral PE of 56.50 (as provided in information), Splash's share price could be (56.5 x 6.53) = 367.

Based on average PE of comparable companies, suggestive share price for Splash is (34.11 x 6.53) = 223.

This indicates that Splash could issue their share much higher than 20 taka per share including 100% premium. Therefore, it can be concluded that 100% share premium expectation by Splash's management could be approved by BSEC as it is justifiable based on NAV, EPS and sectoral PE.

6.2. Maximum premium can be charged if board expected premium is not approved:

Based on NAV, EPS and Sectoral PE, it is more likely that BSEC will approve 100% premium on 10 Taka face value. However, Splash's expected share premium is too low. Comparing the NAV, EPS and PE of comparable companies, Splash could issue share at a price ranging from 123 taka per share to 367 taka per share. Therefore, Splash could raise expected fees up to 1000% premium considering its NAV and EPS.

6.3. Number of shares to be issued in IPO considering issue cost and premium permissible:

For the selected option, cash flow in 5.1 shows that Splash will require BDT 120,000,000 in the 1st year, BDT 20,000,000 in the second year and BDT 6,000,000 in the third year. That means total fund requirement is BDT 146,000,000 for three years. According to the provided information, there is 3% issue cost involved. Considering this, total fund needs to be raised from IPO is 146,000,000/0.97 = 150,515,464. So total number of new shares to be issued = 150,515,464/20 = 7,525,773.

6.4. Conclusion:

- The proposed 100% share premium (10 Taka on a face value of 10 Taka) by Splash's management is supported by higher-than-average NAV (39.38 vs. 36.4) and EPS (6.53), making the premium reasonable when compared to industry peers.
- Calculations based on NAV, EPS, and sectoral PE suggest Splash's share price could range from 123 to 367 Taka, which justifies the premium.
- Splash's share price, based on sectoral PE, could be significantly higher than 20 Taka (including the 100% premium), indicating that Splash could potentially raise a much larger premium, up to 1000%, if desired.
- Given that the 100% premium is well-supported by key financial metrics, it is likely that BSEC will approve Splash's proposed premium based on NAV, EPS, and sectoral PE.
- Splash requires 146 million BDT over three years, and after accounting for a 3% issue cost, the total fund to be raised is 150.52 million BDT, necessitating the issuance of 7,525,773 new shares at the proposed 20 Taka per share.
- During initial 5 years, no new bank loan can be availed without the consent of new shareholders and the BSEC

6.5. Recommendation:

- Splash should consider increasing its proposed premium as financial analysis suggests it could charge a much higher premium, potentially up to 1000%, given its NAV, EPS, and sectoral PE.
- Even though the 100% share premium is justifiable, Splash should prepare for potential negotiations with BSEC and be ready to provide detailed justifications based on the favourable financial comparisons to industry peers.

- Given the fund requirement over the next three years and the total number of shares needed, Splash should plan its IPO share issuance carefully to balance raising adequate capital while maintaining shareholder value.
- Splash should also consider the market sentiment and potential investor appetite when setting the final share price to ensure a successful IPO.
- Splash management should review the authenticity and accuracy of growth forecast for revenue and cost.

7. Comment on suitability of financing through bank loan: (Requirement b-iv):

7.1. Financing through bank loan:

As financing strategy, Splash management is also considering feasibility of financing the electrolyte drinking line project through bank loan. Bank loan is available for Splash at Six Month Moving Average Rate of Treasury Bill (SMART) rate plus 3.5% premium. Current SMART is 9.5% which means current borrowing rate will be 9.5% + 3.5% = 13%. The borrowing rate is lower than the CAPM rate of 14% require by Splash management.

Electrolyte drink project has positive NPV after discounting the Free Cash Flows at 14%. If the project is financed by bank loan, cost of capital will become 13% which is lower than the 14% CAPM rate. Therefore, NPV of the project will be even higher. This means financing with bank loan will also be feasible.

7.2. Conclusion:

- Splash board is considering bank loans as an alternative to financing through IPO.
- If Splash management plans to finance the project through bank loan, cost of capital would be 13%.
- Evaluation of the project feasibility with cost of bank loan would generate higher NPV.

7.3. Recommendation:

- Financing the electrolyte drink project through a bank loan is feasible and advantageous.
- Lower borrowing rate compared to the CAPM rate will result in a higher NPV, further enhancing the project's profitability.
- Board may consider negotiating with bank to reduce the premium over SMART rate. Reduction in premium will reduce the borrowing rate and lead to higher profitability.

8. Comment on ethical issues of Splash's business and operations (Requirement C-i).

8.1. Ethical issues:

Ethical issues involving Splash's business and operation includes:

Extraction of underground fresh water:

Splash and other water bottle companies mostly extracts underground fresh waters and sell them to the market. These underground waters should benefit or used by all people. But the bottled water industry extracts underground water (with or without licenses) for their commercial purposes and enjoys the benefit. Furthermore, to maximize profit, they try to extract as long as possible. These leads to over extraction of limited underground water. Other members of the society do not get any benefit from this. Because of over extraction of underground water, risk of land slide has been increase significantly in recent years.

Appointment of auditor at lower fees:

A. Chowdhury & Co., Chartered Accountants (ACC) has been group auditor for several years. Splash has appointed ACC as its auditor at a fee much lower than previous auditor. This indicates that ACC has been appointed due to their undercutting of fees. Furthermore, ACC have been banned by Bangladesh Securities & Exchange Commission (BSEC) for collaborating with an asset management company to embezzle investors' fund. Splash may have known this fact at the time of appointment. This raises question on ethical behaviour for both ACC and Splash.

Increasing price of bottled water:

Water is very essential for survival of lives. Due to lack of safe water sources and convenience, popularity of bottled water has been increasing. However, with the rise of demand, the price of the for bottled water has been increasing. Manufacturers of the bottled water are exploiting the increasing demand with price increase.

Colluding with municipal water supply:

One of the reasons for increasing demand of bottled water is the unsafe drinking water supplied by the municipality. There are chances that bottled water manufacturers collude with the municipality officials in order to keep supplying water unsafe for drinking. As a result, demand for bottled water keep raising.

8.2. Conclusion

- Splash and similar bottled water companies over-extract underground fresh water, a limited natural resource, for benefits while depriving society of equitable access to these resources.
- Splash's appointment of A. Chowdhury & Co. (ACC) as its auditor at significantly lower fees, despite the firm's questionable reputation and previous regulatory issues, suggests a potential compromise in ethical standards.
- The increasing popularity of bottled water, due to a lack of safe drinking water alternatives, has been met with rising prices. Bottled water manufacturers, including Splash, are exploiting consumer demand to maximize profits.

- There is a potential conflict of interest where bottled water manufacturers might collude with municipal authorities to maintain unsafe water supplies, increasing dependence on bottled water.

8.3. Recommendation

- Splash should adopt sustainable water extraction practices, ensuring that underground water resources are conserved and available for the benefit of the wider community.
- Splash should prioritize transparency and integrity by appointing auditors with a clean and ethical track record.
- Splash should reconsider its pricing strategy for bottled water to avoid exploiting vulnerable consumers who lack access to safe drinking water.
- Splash should proactively distance itself from possible collusion with authorities and support efforts to improve municipal water safety, which would reduce unethical profiteering at the expense of public health.

9. Assessment of sustainability factors of Splash's business (Requirement C-ii).

A business can earn sustainability when it achieves Human Sustainability, Social Sustainability, Economic Sustainability and Environmental Sustainability.

9.1. Human Sustainability:

Human sustainability focuses on maintaining and improving the human capital in society by investments in the health and education systems, access to services, nutrition, knowledge and skills. Following human sustainability factors has been identified from Splash's business operation.

Access to Safe Drinking Water:

Water supplied by the municipal authorities are of poor quality and not safe to drink. Bottled water plays a significant role in providing public access to affordable and reliable water sources. Furthermore, the bottled water provides convenience to the consumers.

Talent Acquisition and Development:

Splash has a HR department that is responsible for recruiting, training, and developing employees across all departments, promoting equal employment opportunities and demographic diversity. It offers specialized training programs that are designed to enhance employee skills, particularly for customer-facing roles, ensuring consistent service quality and customer satisfaction.

Health and lifestyle:

Splash ensures that bottles and the water are safe for human to consume. Bottled water products go through security systems that include controlled access to the plant and critical production areas, gating, motion sensors, electronic contact security alarms, and tamper-evident locks. It is ensured that water intake systems remain sanitary and sealed. The rise of electrolyte drinks also caters to health-conscious consumers, emphasizing hydration and mineral replenishment for active lifestyles.

9.2. Social Sustainability:

Social Sustainability is focused on identifying and managing business impacts, both positive and negative, on people. Social sustainability aims to preserve social capital by investing and creating services to communities and its cultures. Following social sustainability factors has been identified from Splash's business operation.

Corporate Social Responsibility (CSR):

Splash actively engages in CSR initiatives to improve local environmental conditions and support community development. These initiatives contribute to human sustainability by improving the well-being of local populations and helping address environmental challenges, though they may not fully offset the industry's negative impacts.

Employment Opportunities:

Splash recruit and train skilled personnel for various roles within the company, including production, quality control, sales, marketing, and logistics. It ensures availability of sufficient human resource and implements various employee development programs to enhance skills, productivity, diversity, and job satisfaction.

Economic Opportunities and Ethical Dilemmas:

The bottled water industry generates employment and boosts local economies, but this comes with ethical dilemmas. Local communities often face difficult choices between the economic benefits of working for the industry and the environmental costs, such as water depletion and ecosystem degradation, that jeopardize long-term sustainability.

Legal Inequality:

Local communities and environmental activists often lack the financial resources and legal power to challenge multinational companies when disputes arise over water extraction and environmental degradation.

9.3. Economic Sustainability:

Economic sustainability aims to maintain the financial capital and improve the standard of living. In the context of business, it refers to the efficient use of assets to maintain company profitability over time. Following economic sustainability factors has been identified from Splash's business operation.

Waste and Overconsumption:

Bottled water is often not fully consumed, resulting in significant waste of both the packaging and the water itself. This contributes to inefficiencies in resource use and exacerbates environmental damage.

Cost and Profit Disparities:

Bottled water companies often profit from using public tap water or extracting water at low costs, while selling the same water at significantly higher prices, raising ethical questions about the commodification of a basic human necessity.

9.4. Environmental Sustainability:

Environmental sustainability aims to improve human welfare through the protection of natural capital (e.g., land, air, water, minerals etc.). Following environmental sustainability factors has been identified from Splash's business operation.

Water Scarcity and Resource Depletion:
The increased extraction of groundwater by bottled water companies exacerbates water scarcity, which can harm agricultural production and reduce access to fresh water for local communities. Groundwater depletion increases the likelihood of landslides and sinkholes around the world.

Plastic Waste and Pollution:
The vast majority of bottled water is sold in plastic packaging, which is non-degradable and a significant contributor to global plastic pollution. The growing waste from these bottles burdens landfills and contributes to ocean pollution.

Depletion of Natural Resources:
Bottled water companies often extract water from underground reservoirs or public tap water systems, depleting essential freshwater resources. This extraction can reduce water availability for local agriculture and ecosystems, potentially leading to water scarcity.

Carbon Footprint:
The life cycle of bottled water (manufacturing, packaging, transportation, etc.) involves significant use of fossil fuels and contributes to greenhouse gas emissions. Compared to tap water, which is more energy-efficient to distribute, bottled water production is a high-energy, high-emission process.

Biodiversity Loss:
The production of bottled water and plastic bottles can lead to habitat destruction, pollution, and land use changes, resulting in a loss of biodiversity.

9.5. Conclusion

- Positive initiatives include talent development and corporate social responsibility, enhancing human and social sustainability.
- Economic practices raise ethical concerns regarding resource use and pricing strategies.
- Environmental challenges include water depletion, plastic pollution, and significant carbon emissions.
- Addressing these challenges is vital for long-term profitability and community trust.
- Splash shows human sustainability efforts by creating diversity in leadership roles and the talent it acquires for bringing versatile technical knowledge and skill to the organization.
- Splash strictly follows the health & safety policies within the company. It has also developed operating policies to improve operational efficiency.
- It also creates various job opportunities which requires staffing for various positions like production, quality control, sales, marketing, and logistics.

9.6. Recommendation

- Invest in water purification technologies and community initiatives that provide safe drinking water to underserved populations, reducing reliance on bottled water.

- Expand corporate social responsibility initiatives to include partnerships with local communities, ensuring that their needs and concerns regarding water extraction and environmental impacts are prioritized.
- Develop and promote programs aimed at reducing waste associated with bottled water, such as incentivizing recycling or introducing refillable options.
- Reevaluate pricing strategies to ensure fair access to water, considering the ethical implications of commodifying a basic human necessity, and invest in local community development.
- Commit to sustainable sourcing practices that protect local ecosystems and biodiversity and adopt measures to mitigate the carbon footprint associated with bottled water production and distribution.
- Foster open communication with local communities and stakeholders about water extraction practices and environmental impacts, providing avenues for feedback and collaboration to address legal and social inequalities.

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10. Appendices:

10.1. Appendix 1 - Analysis of Adjusted Balance Sheet:

Statement of Financial Position
As of 31 December 2023

	31-Dec-23	Adjustment	Adjusted 31-Dec-2023	31-Dec-22
Assets				
Non-current assets				
Property, plant and equipment	238,494,498	(6,000,000)	232,494,498	227,618,158
Intangible Assets	19,477,584	-	19,477,584	21,466,428
Right-of-use assets	193,338,984	-	193,338,984	206,607,900
Other financial assets	216,288,553	(50,000,000)	166,288,553	156,304,260
Non-current advance	-	13,333,333	13,333,333	
Deferred tax asset	42,853,988	-	42,853,988	60,799,164
	710,453,607	**(42,666,667)**	**667,786,940**	**672,795,910**
Current assets				
Inventories	10,126,848	-	10,126,848	6,246,972
Trade and other receivables	8,711,606	-	8,711,606	4,144,812
Advances, deposits and prepayments	19,878,180	(3,333,333)	16,544,847	9,199,572
Cash and bank equivalents	16,455,672	-	16,455,672	9,276,864
	55,172,306	**(3,333,333)**	**51,838,973**	**28,868,220**
Total assets	**765,625,913**	**(46,000,000)**	**719,625,913**	**701,664,130**
EQUITY AND LIABILITIES				
Capital and reserves				
Share capital	194,415,000	-	194,415,000	194,415,000
Retained earnings	269,599,200	(46,306,950)	223,292,250	142,701,175
Additional equity contribution		638,847	638,847	-
	464,014,200	**(45,668,103)**	**418,346,097**	**337,116,175**
Non-current Liability				
Long term borrowings	55,847,147	(331,897)	55,515,250	112,940,969
Lease Liabilities	80,077,020	-	80,077,020	114,509,010
Post-employment benefit obligations	14,747,496	-	14,747,496	10,525,247
	150,671,663	**(331,897)**	**150,339,766**	**237,975,226**
Current liabilities				
Lease Liabilities-current portion	34,441,224	-	34,441,224	34,441,224
Short term borrowings	26,399,410	-	26,399,410	26,950,968
Trade and other payables	38,827,488	-	38,827,488	36,016,590
Income tax provision	51,271,928	-	51,271,928	29,163,947
	150,940,050	**-**	**150,940,050**	**126,572,729**
Total liabilities	**301,611,713**	**(331,897)**	**301,279,816**	**364,547,955**
Total equity and liabilities	**765,625,913**	**(46,000,000)**	**719,625,913**	**701,664,130**

10.2. Appendix 2 - Analysis of Adjusted Statement of Profit & Loss:

	31-Dec-23	Adjustment	Adjusted 31-Dec-2023	31-Dec-22	Change	Change %
Revenue	963,116,460	-	963,116,460	690,703,200	272,413,260	39.4%
Cost of Revenue	(386,232,000)	-	(386,232,000)	(319,128,750)	(67,103,250)	21.0%
Gross profit	**576,884,460**	**-**	**576,884,460**	**371,574,450**	**205,310,010**	**55.3%**
Administrative expenses	(350,805,360)	(66,000,000)	(416,805,360)	(222,945,695)	(193,859,665)	87.0%
Selling & Ad expenses	(53,952,495)	20,000,000	(33,952,495)	(44,588,934)	10,636,439	-23.9%
Finance costs	(27,905,148)	(306,950)	(28,212,098)	(29,004,564)	792,466	-2.7%
Income from operation	**144,221,457**	**(46,306,950)**	**97,914,507**	**75,035,257**	**22,879,250**	**30.5%**
Non-operating income	16,003,320	-	16,003,320	8,290,308	7,713,012	93.0%
Profit before tax	**160,224,777**	**(46,306,950)**	**113,917,827**	**83,325,565**	**30,592,262**	**36.7%**
Current tax (expense)/income	(51,271,928)	-	(39,871,239)	(29,163,947)	(10,707,292)	36.7%
Deferred tax (expense)/income	17,945,176	-	9,967,810	7,290,987	2,676,823	36.7%
Profit for the year	**126,898,025**	**(46,306,950)**	**84,014,398**	**61,452,605**	**22,561,793**	**36.7%**
Other comprehensive income	-	-	-	-	-	0.0%
Total Comprehensive Income	**126,898,025**	**(46,306,950)**	**84,014,398**	**61,452,605**	**22,561,793**	**36.7%**

10.3. Appendix 3 - Mix analysis:

Revenue	**Contribution**	
	2023	**2022**
Sell to DDL	81.0%	86.5%
Direct sells to corporates	15.0%	11.0%
Private level sells	4.0%	2.5%
	100%	**120%**
Cost of Revenue		
Prime cost	38.0%	41.0%
Indirect cost	62.0%	59.0%
	100%	**58%**

10.4. Appendix 4: Ratio Analysis:

SL.	Ratio Name	31-Dec-23	31-Dec-22
1	Current Ratio	0.34	0.23
2	Quick Ratio	0.28	0.18
3	Debt to equity ratio	0.72	1.08
4	Asset Turnover Ratio	1.34	0.98
5	TIE Ratio	5.0	3.9
6	Return on Asset	11.7%	8.8%
7	Return on Equity	20.1%	18.2%

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